

MODULE 1 INTEGUMENTARY

The Integumentary System.

INTEGUMENTARY & SKIN

The skin is the largest organ in the body and the first place all four tissue types show up working together. It is also the organ you can read from across the room, which makes it the best teacher of the structure to consequence habit.

COMPETENCIES W1-SKIN-LAYERS, W1-EPIDERMAL-STRATA, W1-DERMIS-LAYERS, W1-SKIN-ACCESSORY

BY THE END

1. Name the components and the six functions of the integumentary system.
2. List the strata of the epidermis deep to superficial and say which is missing in thin skin.
3. Name the four principal cells of the epidermis and what each does.
4. Describe melanin transfer and why it protects the nucleus.
5. Separate the papillary from the reticular region of the dermis.
6. Describe hair anatomy, the follicle, and the bulb.
7. Compare the four skin glands on distribution, secretion, and onset.
8. Name the components of a nail and their functions.
9. Read skin color as a diagnostic clue.



WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Integumentary

drsrennie-stack.github.io/new-build-bio4-solano/integumentary-concept-videos.html

Components and Functions

Components. The skin, also called the cutaneous membrane, plus associated structures: hair, oil glands, sweat glands, sensory receptors, and nails.

Dermatology is the medical specialty dealing with the structure, function, and disorders of the integumentary system.

THE SIX FUNCTIONS OF THE SKIN

Function	How the structure delivers it
Thermoregulation	Sweat glands and dermal blood vessels adjust heat loss
Blood reservoir	Extensive dermal vascular network holds a substantial volume of blood
Protection	Keratin, lamellar granules, melanin, and immune cells
Cutaneous sensations	Tactile discs, corpuscles of touch, hair root plexuses, free nerve endings
Excretion and absorption	Small amounts of waste out, lipid-soluble substances in
Synthesis of vitamin D	UV exposure begins the synthesis pathway in the skin

General information. The skin is the largest organ by weight, covering the external surface. In an adult the area is roughly 22 square feet, and it weighs 10 to 11 pounds, about 7 percent of total body weight.

The Three Layers

1. **Epidermis**, part of the skin.
2. **Dermis**, part of the skin.
3. **Subcutaneous tissue**, not part of the skin.

The Epidermis

The thinner, superficial portion, composed of epithelial tissue and avascular. Its composition is **keratinized stratified squamous epithelium**. Avascular matters: the epidermis depends entirely on diffusion from the dermis beneath it, which is why a deep burn that destroys dermal vessels cannot support regeneration.

The four principal cells

CELLS OF THE EPIDERMIS		
Cell	Share and origin	Function
Keratinocytes	90 percent, arranged in 4 to 5 layers	Produce keratin, a tough fibrous protein protecting skin and underlying tissue from abrasion, heat, microbes, and chemicals. Also produce lamellar granules, which release a water-repellent sealant that decreases water entry and loss and inhibits entry of foreign material
Melanocytes	8 percent, develop from the ectoderm of the embryo	Produce melanin, a yellow-red or brown-black pigment contributing to skin color and absorbing damaging UV light
Intraepidermal macrophages (Langerhans cells)	Originate in red bone marrow	Migrate to the epidermis and participate in immune responses against microbes invading the skin. Help other immune cells recognize and destroy an invader. Easily damaged by UV light
Merkel cells (tactile cells)	Least numerous, located in the deepest layer of the epidermis	Contact the flattened process of a sensory neuron, called a Merkel disc or tactile disc

The strata

Most regions of the body have **four strata**: stratum basale, stratum spinosum, stratum granulosum, stratum corneum. Areas of greatest friction, meaning fingertips, palms, and soles, have **five strata**, adding the stratum lucidum between the granulosum and the corneum.

MNEMONIC

Baby Screams Grandpa Loves Coffee. Basale, Spinosum, Granulosum, Lucidum, Corneum. Deep to superficial.

Melanin transfer

1. Melanocytes have long slender projections that extend between keratinocytes and transfer melanin granules to them.

2. Once inside the keratinocyte, melanin granules cluster to form a protective veil over the nucleus, on the side facing the skin surface.
3. That veil shields the DNA, preventing UV damage from the sun.
4. Melanocytes themselves remain susceptible to damage by UV light. They protect everyone else and not themselves.

The Dermis

The second, deeper portion of the skin. **Tissue composition.** Dense irregular connective tissue with collagen and elastic fibers.

Functions. Gives the skin great tensile strength, and allows stretch and recoil.

Characteristics. Thicker than the epidermis and variable by body site, greatest at the soles and palms. Contains blood vessels, nerves, glands, and hair follicles. **Cells.** Fibroblasts, a few macrophages, and a few adipocytes near the boundary with the subcutaneous layer. The dermis is essential to the survival of the epidermis and its associated structures.

THE TWO PRINCIPAL LAYERS OF THE DERMIS		
	Papillary region	Reticular region
Position and size	Thinner, superficial. One fifth of the total dermal layer	Thicker, deep. Four fifths of the total dermal layer
Contents	Thin collagen and fine elastic fibers. Dermal papillae, small finger-shaped projections into the undersurface of the epidermis	Bundles of thick collagen fibers, scattered fibroblasts, wandering macrophages, adipocytes in the deepest part with coarse elastic fibers
Structures within	Capillary loops, Meissner corpuscles for touch and pressure, free nerve endings for temperature, light touch, and itching	Blood vessels, nerves, hair follicles, sudoriferous sweat glands, sebaceous glands
Function	Anchors epidermis, supplies it by diffusion, carries the fine touch receptors	Net-like layer attached to the subcutaneous layer. Provides strength, extensibility, and elasticity. The regular net-like collagen orientation makes skin resistant to stretching

Epidermal ridges, the fingerprints

The bond between dermis and epidermis on the surfaces of the palms, soles, and fingertips. **Functions.** Increase surface area, which increases the number of corpuscles of touch and therefore tactile sensitivity, and increase grip by increasing friction.

The Subcutaneous Layer (Hypodermis)

Deep to the dermis and not part of the skin. **Composition.** Areolar and adipose tissue. Fibers extend from the dermis and anchor the skin to this layer and to the fascia around muscles and bones. **Function.** Storage for fat. **Associated components.** Lamellated corpuscles, which are nerve endings sensitive to pressure, and large blood vessels.

Skin Color

THREE PIGMENTS		
Pigment	Color contributed	Notes
Melanin	Pale yellow to reddish-brown	Two forms: pheomelanin, yellow to red, and eumelanin, brown to black
Hemoglobin	Pink to red	Seen through the epidermis, depends on the oxygen content of blood in the dermal capillaries
Carotene	Yellow-orange	A precursor to vitamin A, used to synthesize pigments for vision

Melanocyte distribution. Most numerous in the epidermis of darker areas such as the nipples and areolae, and in the face, limbs, and mucous membranes. The number of melanocytes is about the same in all people. The difference in skin color comes from the amount of pigment those cells produce and transfer to keratinocytes.

Melanin synthesis. Made from the amino acid tyrosine in the presence of the enzyme tyrosinase, inside an organelle called the **melanosome**. Exposure to UV light increases the activity of that enzyme, which increases melanin production. That is a tan, and it is protective against further exposure.

Hair

Location. Most skin surfaces. Exceptions: palms, fingertips, soles, and plantar surfaces of the feet.

Determinants of thickness and distribution. Genetic and hormonal.

Functions. Limited protection from the sun, decreased heat loss from the scalp, eyelashes and eyebrows protecting the eyes from debris, nostril and ear hair protecting those structures from debris, and sensing light touch via hair root plexuses.

Anatomy of a hair

Composition. Dead, keratinized epidermal cells. **Two parts.** The **hair root**, the portion penetrating into the dermis and sometimes the subcutaneous tissue, and the **hair shaft**, the portion above the surface of the skin. Both have the same three concentric layers.

1. **Medulla**, the inner layer, 2 to 3 rows of irregular cells containing pigment.
2. **Cortex**, the major part of the shaft, elongated cells.
3. **Cuticle**, the outermost, a single layer of thin, flat, heavily keratinized cells.

The hair follicle

The follicle surrounds the root of the hair.

- **External root sheath**, a downward continuation of the epidermis.
- **Internal root sheath**, produced by the matrix, forming a cellular tubular sheath of epithelium. External plus internal root sheath equals the **epithelial root sheath**.
- **Dermal root sheath**, the dermis surrounding the hair follicle.

The bulb is the onion-shaped structure at the base of each follicle, surrounding the dermal root sheath. It contains the **papilla of the hair**, which is areolar connective tissue with many blood vessels, and the **hair matrix**, the germinal layer of cells arising from the stratum basale. The matrix is the site of cell division: it is responsible for the growth of existing hairs, the generation of new hairs when old ones are shed within the same follicle, and it gives rise to the cells of the internal root sheath.

Structures associated with hair

- **Sebaceous oil glands**, secrete oil onto the hair.
- **Arrector pili muscles**, smooth muscle that pulls the hair shaft perpendicular, producing goose bumps.
- **Hair root plexus**, dendrites of neurons surrounding each follicle, detecting light touch.

Types of hair and hair color

Lanugo is the very fine, non-pigmented downy hair covering the fetus. **Terminal hairs** are long, coarse, heavily pigmented hairs that replace lanugo just before birth on the eyebrows, eyelashes, and scalp. **Vellus hairs**, the peach fuzz, replace lanugo on the rest of the body. After puberty, under hormonal influence, some vellus hairs are replaced by terminal hairs, and the proportions differ between individuals.

Hair color is determined by the amount and type of melanin in the keratinized cells. Melanin is synthesized by melanocytes, scattered in the matrix of the bulb, and passes into the cells of the cortex and medulla. Dark hair is mostly eumelanin. Blond and red hair are mostly pheomelanin. Gray hair reflects a progressive decline in melanin production and contains few melanin granules. White hair lacks melanin entirely and accumulates air bubbles in the shaft.

Skin Glands

THE FOUR GLANDS COMPARED					
Gland	Distribution	Secretory portion	Duct terminates	Secretion	Onset
Sebaceous (oil)	Largely lips, glans penis, labia minora, tarsal glands. Small on trunk and limbs. Absent on palms and soles	Dermis	Mostly connected to a hair follicle	Sebum, a mixture of triglycerides, cholesterol, proteins, and inorganic salts	Relatively inactive through childhood, activated at puberty
Eccrine sweat	Throughout most regions of the body, especially forehead, palms, and soles	Mostly deep dermis, sometimes the upper subcutaneous layer	Surface of the epidermis	Perspiration: water, ions (sodium, chloride), urea, uric acid, ammonia, amino acids, glucose, lactic acid	Soon after birth
Apocrine sweat	Skin of axillae, groin, areolae, bearded regions of the face, clitoris and labia minora	Mostly deep dermis and upper subcutaneous layer	Hair follicles	Perspiration with the same components as eccrine sweat plus lipids and proteins	Puberty
Ceruminous	External auditory canal	Subcutaneous layer	Surface of the external auditory canal, or into ducts of sebaceous glands	Cerumen, a waxy material	Soon after birth

Nails

Definition. Plates of tightly packed, hard, dead keratinized epidermal cells forming a clear, solid covering over the dorsal surfaces of the distal portions of the digits.

COMPONENTS OF A NAIL

Component	What it is
Nail body (plate)	The visible portion, comparable to the stratum corneum of the skin except the flattened keratinized cells fill with a harder keratin and are not shed. Mostly pink due to the vascularity of the tissue below
Free edge	The part extending beyond the finger or toe. White because there are no capillaries beneath it
Nail root	The portion buried in a fold of skin, not visible
Lunula	The little moon, a crescent-shaped area at the proximal end of the nail body. Vascular tissue does not show through because the epithelium there is thickened
Hyponychium	Thickened stratum corneum below the free edge. Secures the nail to the fingertip
Eponychium (cuticle)	A narrow band of epidermis extending from and adhering to the margin of the nail wall. It is stratum corneum
Nail matrix	Epithelium proximal to the nail root, containing dividing cells that produce new nail cells

Functions. Protect the ends of the digits, provide support and counter pressure to the palmar surface of the fingers which enhances touch perception and manipulation, and allow us to grasp and manipulate small objects, scratch, and groom.

Thick Skin and Thin Skin

The greatest contributor to epidermal thickness is an increased number of layers in the stratum corneum, which arises in response to greater mechanical stress.

STRUCTURAL DIFFERENCES

Feature	Thin (hairy) skin	Thick (hairless) skin
Distribution	All parts of the body except palms, palmar surface of digits, and soles	Palms, soles, palmar surface of digits
Epidermal strata	Stratum lucidum essentially lacking, thinner spinosum and corneum	Stratum lucidum present, thicker spinosum and corneum
Epidermal ridges	Lacking, because dermal papillae are poorly developed, fewer, and less organized	Present, because dermal papillae are well developed, more numerous, and organized in parallel rows
Hair follicles and arrector pili	Present	Absent
Sebaceous glands	Present	Absent
Sudoriferous glands	Fewer	More numerous
Sensory receptors	Sparser	Denser

Immune Protection, Sensation, Exchange

Two cell types with an immunological role

1. **Intraepidermal macrophages (Langerhans cells)**, alert the immune system to potentially harmful microbial invaders, recognizing and processing them.
2. **Macrophages in the dermis**, phagocytize bacteria and viruses that get past the intraepidermal macrophages of the epidermis.

Cutaneous sensations

Sensations that arise in the skin, including tactile sensations: touch, pressure, vibration, tickling, and thermal sensation. **Pain** is the sensation indicating impending or actual tissue damage. The receptors include tactile discs of the epidermis, corpuscles of touch, and hair root plexuses around each follicle.

Excretion and absorption

The skin has a small role in both. **Excretion** is the elimination of substances from the body. **Absorption** is the passage of materials from the external surface inward, and although the amount is negligible, it does occur.

What crosses. Lipid-soluble materials: the fat-soluble vitamins A, D, E, and K, certain drugs, gases such as oxygen and carbon dioxide, toxic materials, salts of heavy metals such as lead, mercury, and arsenic, and the oils of poison ivy and poison oak. Steroid drugs such as cortisone are lipid soluble and move into the papillary region of the dermis, where their anti-inflammatory effect includes inhibiting histamine production by mast cells. Other drugs are delivered through patches, nitroglycerin and fentanyl among them.

Synthesis of vitamin D

Only a small amount of exposure, 10 to 15 minutes twice a week, is necessary for vitamin D synthesis.

Clinical Correlations

Skin grafts

A **skin graft** is the transfer of a patch of healthy skin from a donor site to cover a wound. New skin cannot regenerate if the injury has damaged a large area of the stratum basale and its stem cells, which is exactly why the depth of a burn determines the treatment.

Benefits. Protects against fluid loss and infection, promotes tissue healing, reduces scar formation, prevents loss of function, and improves cosmetic outcome.

Complication. Tissue rejection. To minimize the risk, transplanted skin is taken from the same person, an **autograft**, or from an identical twin, an **isograft**. In severe burns where taking an autograft would cause more harm than good, **autologous skin transplantation** is used: small amounts of the person's epidermis are removed, keratinocytes are cultured in the lab and grown into thin sheets of skin, and those sheets are transplanted.

Albinism and vitiligo

Albinism is the inherited inability to produce melanin. Most affected individuals have melanocytes but cannot synthesize tyrosinase, so melanin is missing from skin, eyes, and hair. Manifestations include problems with night vision and a tendency for the skin to burn with sun overexposure.

Vitiligo is the partial or complete loss of melanocytes from patches of skin, producing irregular white spots.

Skin color as a diagnostic clue

FOUR COLOR CHANGES AND WHAT THEY SUGGEST

Finding	Appearance	What it suggests
Cyanosis	Blue	Blood is not picking up adequate oxygen from the lungs. Most apparent in mucous membranes, nail beds, and skin
Jaundice	Yellowish skin and sclera	A build-up of the yellow pigment bilirubin, likely indicating liver disease
Erythema	Redness	Engorgement of dermal capillaries with blood, from skin injury, heat exposure, infection, inflammation, or allergic reaction
Pallor	Paleness	May occur in conditions such as shock or anemia

These changes can be difficult to assess in individuals with darker skin. Examining the nail beds, gums, mucous membranes, and conjunctiva is the reliable approach.

Thermoregulation

THE SKIN'S TWO-DIRECTION RESPONSE

When the body is	Blood vessels	Sweat	Other
Too warm	Vasodilation brings warm blood to the surface so heat is lost to the environment	Sweat production increases, and evaporation cools the surface	Goal is to shed heat
Too cold	Vasoconstriction keeps blood away from the surface so heat is conserved in the core	Sweat production is reduced	Shivering generates heat through muscle activity

THE SKIN LAYERS MAP THE DEPTH OF A BURN

Burn classification is anatomy applied. A superficial burn involves the epidermis only, and it heals because the stratum basale is intact and its stem cells can repopulate the layers above. A partial-thickness burn reaches into the dermis, blisters because the dermal-epidermal junction fails, and still heals from surviving follicles and glands, since those structures are lined with epidermal cells that live deep in the dermis. A full-thickness burn destroys the dermis and its appendages, so there is no source of new epidermis left within the wound, and grafting becomes necessary. Read every burn question by asking one thing first: how deep did it go, and is the stratum basale still there.